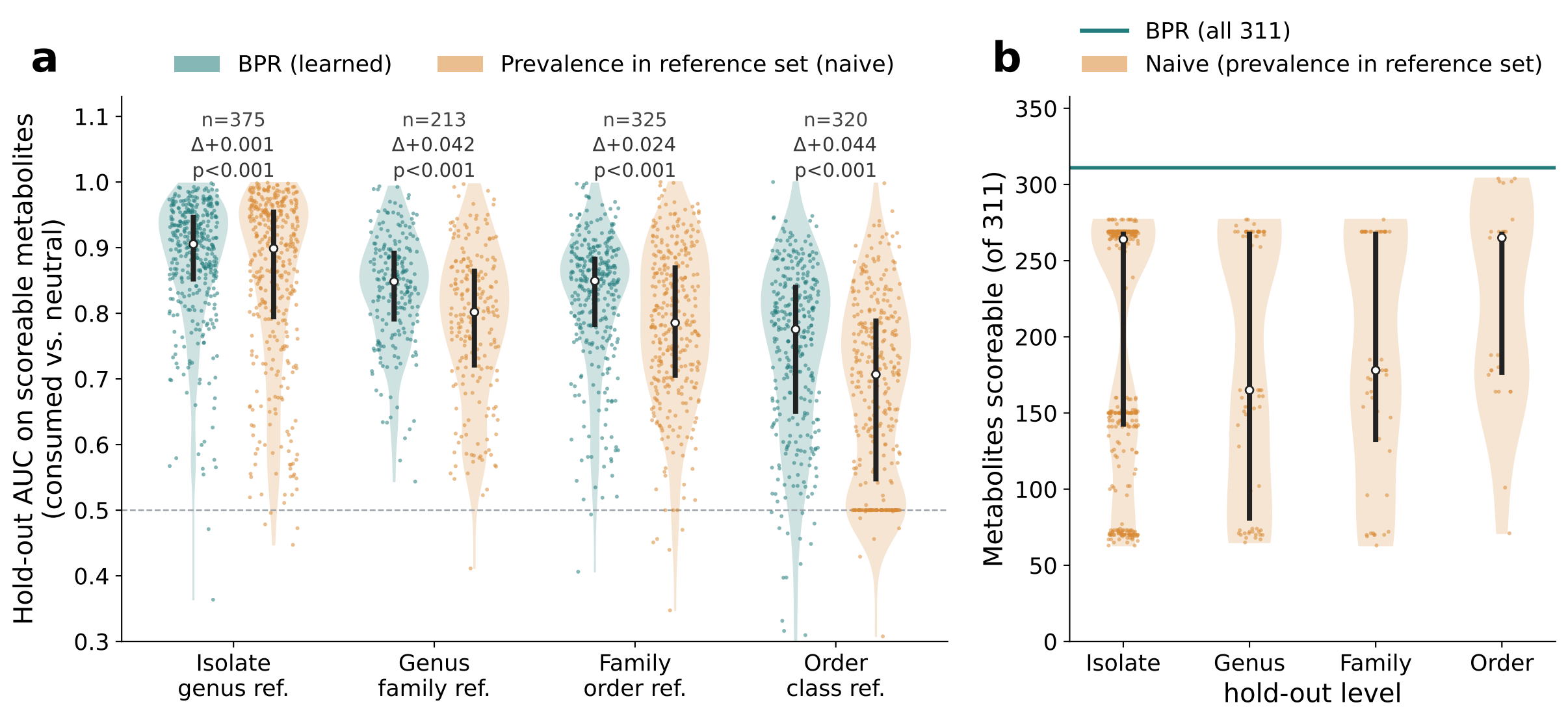

a
BPR (learned)
Prevalence in reference set (naive)
Hold-out AUC on scoreable metabolites
(consumed vs. neutral)
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
n=375
Δ+0.001
p<0.001
n=213
Δ+0.042
p<0.001
n=325
Δ+0.024
p<0.001
n=320
Δ+0.044
p<0.001
Isolate
genus ref.
Genus
family ref.
Family
order ref.
Order
class ref.
b
BPR (all 311)
Naive (prevalence in reference set)
Metabolites scoreable (of 311)
350
300
250
200
150
100
50
0
Isolate
Genus
Family
Order
hold-out level